

23/24 Sem 1 Q2(c)

$$(i) \lim_{n \rightarrow \infty} \left(\frac{3n+2}{3n-1} \right)^n \rightarrow \textcircled{1} e^{\ln} \text{ trick: } \left(\frac{3n+2}{3n-1} \right)^n \\ = e^{\ln \left(\frac{3n+2}{3n-1} \right)^n} \\ = e^{n \ln \left(\frac{3n+2}{3n-1} \right)}$$

$\searrow$

$$\textcircled{2} \boxed{\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n} \right)^n = e^x}$$

$$\lim_{n \rightarrow \infty} \left(\frac{3n+2}{3n-1} \right)^n = \lim_{n \rightarrow \infty} \left(\frac{3n+2}{3n-1} \cdot \frac{1}{\frac{1}{3n}} \right)^n \\ = \lim_{n \rightarrow \infty} \left(\frac{1 + \frac{2}{3n}}{1 - \frac{1}{3n}} \right)^n \\ = \lim_{n \rightarrow \infty} \left(\frac{1 + \frac{2/3}{n}}{1 + \frac{(-1/3)}{n}} \right)^n \\ = \lim_{n \rightarrow \infty} \frac{\left(1 + \frac{2/3}{n} \right)^n}{\left(1 + \frac{(-1/3)}{n} \right)^n} \\ = \frac{e^{2/3}}{e^{-1/3}} \\ = e$$

$$(ii) \lim_{n \rightarrow \infty} \sqrt{n} \sin \frac{1}{\sqrt{n}} = \lim_{n \rightarrow \infty} \frac{\sin \frac{1}{\sqrt{n}}}{\frac{1}{\sqrt{n}}}$$

$$\text{Let } \ell = \frac{1}{\sqrt{n}}$$

$$\text{As } n \rightarrow \infty, \ell \rightarrow 0^+$$

Facts about sin

$$(1) -1 \leq \sin \theta \leq 1$$

$$(2) \boxed{\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1}$$

$$= \lim_{\ell \rightarrow 0^+} \frac{\sin \ell}{\ell}$$

$$= 1$$